

Markov Conditions and Factorization in Logical Credal Networks¹

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Abstract

We examine the recently proposed language of *Logical Credal Networks*, a powerful representation formalism that combines probabilities and logic. In particular we investigate the consequences of distinct Markov conditions upon their underlying semantics. We introduce the notion of structure for a Logical Credal Network and show that a structure without directed cycles leads to a well-known factorization result. For networks with directed cycles, we discuss the differences between Markov conditions, factorization results, and specification requirements. We consider several scenarios in causal reasoning that can be tackled by the formalism, in particular looking at partial identifiability and cycles.

Keywords: Logical credal networks, probabilistic logic, Markov condition, factorization, causal reasoning.

1. Introduction

This paper examines *Logical Credal Networks*, a formalism recently introduced by Marinescu et al. [22, 23] to combine logical sentences, probabilities and independence relations. They have proposed a number of novel ideas and evaluated the formalism in practical scenarios with positive results.

The central element of a Logical Credal Network (LCN) is a collection of constraints over probabilities that are written, roughly speaking, as $\alpha \leq \mathbb{P}(\phi|\varphi) \leq \beta$, where ϕ and φ are logical sentences. Independence relations are then extracted

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mostly from the logical content of those inequalities. This scheme differs from previous proposals that extract independence relations from explicitly specified graphs [1, 9, 10], or from explicit syntactic constructs [3, 11, 15].

While Logical Credal Networks have points in common with existing formalisms, they do carry new machinery that deserves attention. For one thing, they feature directed graphs that may contain directed cycles. Cycles occur in important scenarios including grammars, common concepts such as “grand-parent” which require recursive definitions, and physical systems that contain feedback loops. Also they are endowed with a sensible Markov condition that is distinct from those previously established for graphical models such as Bayesian networks [30], Markov networks [19] or credal networks [8]. Little is known about the consequences of these features, and how they interact with the syntactic conventions that turn logical formulas into edges in graphs. In particular, no study has focused on the consequences of those (and other) Markov conditions on factorization results; that is, how such conditions affect the factors that constitute probability distributions.

In this paper we seek a deeper understanding of Logical Credal Networks, by looking at their semantics, their Markov conditions, [and](#) their factorization properties. Conceptually, our goal is rather sharp: to verify the connection between Markov conditions and factorization in the context of propositional Logical Credal Networks (in this paper we do not deal with first-order formulas). As escorting goals, we look into the meaning of directed, bi-directed and undirected edges, and we investigate how to better support causal inference with the flexibility afforded by Logical Credal Networks.

We summarize necessary background in Sections 2 and 3. We introduce the notion of “structure” for an LCN in Section 4. We then show, in Section 5, that the local Markov condition proposed by Marinescu et al. [22] collapses to the usual local Markov condition applied to chain graphs when the structure has no directed cycles. We analyze the behavior of the former Markov condition in the presence of directed cycles, in particular investigating factorization properties and discussing various possible semantics, in Section 6. We then examine practical examples of inference with Logical Credal Networks, in particular looking at causal reasoning in Section 7. Finally, Section 8 provides concluding remarks and highlights several directions of future work.

2. Graphs and Markov Conditions

In this section, we present the necessary concepts related to graphs and graph-theoretical probabilistic models (Bayesian networks, Markov networks, and chain graphs). Definitions and notation vary across the huge literature on these topics; we rely here on three sources. We use definitions by Marinescu et al. [22] and by Spirtes [34] in their work on LCNs and on directed graphs, respectively; we also use standard results from the textbook by Cowell et al. [7].

A *graph* is a triple $(\mathcal{N}, \mathcal{E}_D, \mathcal{E}_U)$, where $\mathcal{N}$ is a set of nodes, and both $\mathcal{E}_D$ and $\mathcal{E}_U$ are sets of edges. A node is always labeled with a proposition, or a logical formula, or a random variable. In fact, we do not distinguish between a

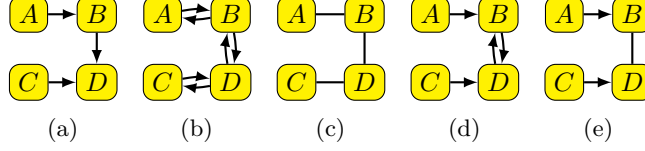


Figure 1: The graphs in Figures 1.a, 1.b, 1.d are directed; the graph in Figure 1.c is undirected; the graph in Figure 1.e is a chain graph.

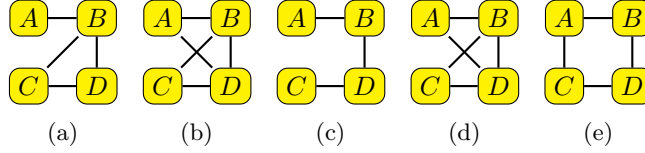


Figure 2: The moral graphs of the graphs in Figure 1.

node and the object associated with it. The elements of $\mathcal{E}_D$ are *directed* edges. A directed edge is an ordered pair of distinct nodes, denoted by $A \rightarrow B$. The elements of $\mathcal{E}_U$ are *undirected* edges. An undirected edge is a pair of distinct nodes, denoted by $A \sim B$; note that nodes are not ordered in an undirected edge, so there is no difference between $A \sim B$ and $B \sim A$. Note that $\mathcal{E}_D$ and $\mathcal{E}_U$ are sets, so there are no multiple copies of elements in them (for instance, there are no multiple undirected edges between two nodes). Note also that there is no loop from a node to itself.

If there is a directed edge from A to B , the edge is said to be *from* A *to* B , and then A is a *parent* of B and B is a *child* of A . The parents of A are denoted by $\text{pa}(A)$. If there are directed edges $A \rightarrow B$ and $B \rightarrow A$ between A and B , we say there is a *bi-directed* edge between A and B and write $A \rightleftarrows B$. If $A \sim B$, then both nodes are said to be *neighbors*. The neighbors of A are denoted by $\text{ne}(A)$. The *boundary* of a node A , denoted by $\text{bd}(A)$, is the set $\text{pa}(A) \cup \text{ne}(A)$. The boundary of a set $\mathcal{B}$ of nodes is $\text{bd}(\mathcal{B}) = \cup_{A \in \mathcal{B}} \text{bd}(A) \setminus \mathcal{B}$. If we have a set $\mathcal{B}$ of nodes such that, for all $A \in \mathcal{B}$, the boundary of A is contained in $\mathcal{B}$, then $\mathcal{B}$ is an *ancestral set*.

A *path* from A to B is a sequence of distinct edges, the first one between A and some node C_1 , then between C_1 and C_2 and so on, until an edge between C_k and B , where all nodes are distinct with the exception that the first and last nodes may be identical, and such that for each pair (D_1, D_2) of consecutive nodes in the path we have either $D_1 \rightarrow D_2$ or $D_1 \sim D_2$ but never $D_2 \rightarrow D_1$. If A and B are in fact identical, the path is a *cycle*. If there is at least one directed edge in a path, the path is a *directed path*; if that path is a cycle, then it is a *directed cycle*. If a path is not directed, then it is *undirected* (as all edges in the path are undirected ones). A *directed/undirected* graph is a graph that only contains directed/undirected edges. A graph without directed cycles is a *chain graph* (note that such a graph may contain undirected cycles). Figure 1 depicts a number of graphs. We have $\text{pa}(B) = \{A\}$ in Figures 1.a and 1.e, $\text{pa}(B) = \{A, D\}$ in Figures 1.b and 1.d, and $\text{pa}(B) = \emptyset$ in Figure 1.c.

If there is a directed path from A to B , then A is an *ancestor* of B and B is a *descendant* of A . For instance, in Figure 1.a, A is the ancestor of B and D is the descendant of B ; in Figure 1.c, there are no ancestors nor descendants of B ; in Figure 1.e, A is the ancestor of B , and there are no descendants of B . As a digression, note that Cowell et al. [7] define “ancestor” and “descendant” somewhat differently, by asking that there is a path from A to B but not from B to A ; this definition is equivalent to the previous one for graphs without directed cycles, but it is different otherwise (for instance, in Figure 1.b the node B has descendants $\{A, B, C, D\}$ in the previous definition but no descendant in the sense of Cowell et al. [7]). Note that there can be no edge from a node to itself, but there can be a cycle from a node to itself, so a node may be its ancestor and its descendant — later the definitions of lcn-descendant and strict descendant offer better concepts in that regard. For now we stick to our former definition, a popular one in the literature [19] that seems appropriate in the presence of directed cycles [34].

We will need the following concepts:

- Suppose we take graph $\mathcal{G}$ and remove its directed edges to obtain an auxiliary undirected graph $\mathcal{G}'$. A nonempty set of nodes $\mathcal{B}$ is a *chain component* of $\mathcal{B}$ iff it is a maximal set of nodes such that any two nodes have a (undirected) path connecting them in $\mathcal{G}'$. Note that a single node left alone in $\mathcal{G}'$ qualifies as a chain component.
- Suppose we take graph $\mathcal{G}$ and add undirected edges between each two nodes that have children in a common chain component of $\mathcal{G}$ and that are not already directly joined in $\mathcal{G}$. Suppose we then take the resulting graph and transform every directed edge into an undirected edge (if $A \rightarrow B$, then both transformed directed edges collapse into $A \sim B$). The final result is the *moral graph* of $\mathcal{G}$, denoted by $\mathcal{G}^m$.
- Suppose we take a graph $\mathcal{G}$ and a set $\mathcal{N}$ of nodes, and we build the moral graph of the smallest ancestral set containing the nodes in $\mathcal{N}$. The resulting graph is denoted by $\mathcal{G}^{ma}(\mathcal{N})$.

Figure 2 depicts the moral graphs that correspond respectively to the five graphs in Figure 1. The moral graph of a directed graph can be produced with relative ease, by noting that in those graphs each node is a chain component. Thus all that must be done is to connect any two nodes that have a common child, and then turn all edges into undirected ones.

There are several formalisms that employ graphs to represent stochastic independence relations. In this paper we focus only on finite spaces, so the concept of stochastic independence is quite simple: random variables X and Y are (conditionally) independent given random variables Z iff $\mathbb{P}(X = x, Y = y | Z = z) = \mathbb{P}(X = x | Z = z) \mathbb{P}(Y = y | Z = z)$ for every possible x and y and every z such that $\mathbb{P}(Z = z) > 0$. In case Z is absent, we have independence of X and Y iff $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y)$ for every possible x and y .

A *Markov condition* explains how to extract independence relations from a graph; there are many such conditions in the literature [7].

Restrict first attention to an undirected graph $\mathcal{G}$ with set of nodes $\mathcal{N}$. The *local Markov condition* for such a graph states that a node A is independent of all nodes in $\mathcal{N}$ other than A itself and A 's neighbors, $\text{ne}(A)$, given $\text{ne}(A)$. The *global Markov condition* states that, given any triple $(\mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_3)$ of disjoint subsets of $\mathcal{N}$, such that $\mathcal{N}_2$ separates $\mathcal{N}_1$ and $\mathcal{N}_3$, then nodes $\mathcal{N}_1$ and $\mathcal{N}_3$ are independent given nodes $\mathcal{N}_2$.² If the probability of every configuration of random variables in $\mathcal{G}$ is larger than zero, then both conditions are equivalent and they are equivalent to a *factorization property*: for each configuration of variables $X = x$, where X denotes the random variables in $\mathcal{G}$, we have $\mathbb{P}(X = x) = \prod_{c \in \mathcal{C}} h_c(x_c)$, where $\mathcal{C}$ is the set of cliques³ of $\mathcal{G}$, each h_c is a function over the random variables in clique c , and x_c is the projection of x over the random variables in clique c .

Now consider a directed acyclic graph $\mathcal{G}$ with a set of nodes $\mathcal{N}$. The *local Markov condition* for such a graph is:

Definition 1 (LMC(D)). A node A is independent, given its parents, of all nodes that are not A itself nor descendants nor parents of A .

The factorization produced by this local Markov condition is

$$\mathbb{P}(X = x) = \prod_{N \in \mathcal{N}} \mathbb{P}(N = x_N | \text{pa}(N) = x_{\text{pa}(N)}), \quad (1)$$

where x_N is the value, in x , of random variable N and $x_{\text{pa}(N)}$ is the projection of x over the parents of N .

Finally, consider a chain graph $\mathcal{G}$ with set of nodes $\mathcal{N}$. The *local Markov condition* for chain graphs is:

Definition 2 (LMC(C)). A node A is independent, given its boundary, of all nodes that are not A itself nor descendants nor boundary nodes of A .

The *global Markov condition* for chain graphs is significantly more complicated:

Definition 3 (GMC(C)). Given any triple $(\mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_3)$ of disjoint subsets of $\mathcal{N}$, if $\mathcal{N}_2$ separates $\mathcal{N}_1$ and $\mathcal{N}_3$ in the graph $\mathcal{G}^{ma}(\mathcal{N}_1 \cup \mathcal{N}_2 \cup \mathcal{N}_3)$, then nodes $\mathcal{N}_1$ and $\mathcal{N}_3$ are independent given nodes $\mathcal{N}_2$.

Assuming the probability of all configurations are positive, then both Markov conditions for chain graphs are equivalent and they are equivalent to a *factorization property*, as follows. Take the chain components $T_1, \dots, T_n$ ordered so that nodes in T_i can only be at the end of directed edges starting from chain components before T_i ; this is always possible in a chain graph. Then the factorization has the form

$$\mathbb{P}(X = x) = \prod_{i=1}^n \mathbb{P}(\mathcal{N}_i = x_{\mathcal{N}_i} \mid \text{bd}(\mathcal{N}_i) = x_{\text{bd}(\mathcal{N}_i)})$$

²In an undirected graph, a set of nodes separates two other sets iff, by deleting the separating nodes, we have no connecting path between a node in one set and a node in the other set.

³A clique is a maximal set of nodes such that each pair of nodes in the set is joined.

where $\mathcal{N}_i$ is the set of nodes in the i th chain component; $x_{\mathcal{N}_i}$ and $x_{\text{bd}(\mathcal{N}_i)}$ are respectively the projection of x over $\mathcal{N}_i$ and $\text{bd}(\mathcal{N}_i)$. Moreover, each factor in the product itself factorizes accordingly to an undirected graph that depends on the corresponding chain component [7]. More precisely, for each chain component T_i , build an undirected graph consisting of the nodes in $\mathcal{N}_i$ and $\text{bd}(\mathcal{N}_i)$ with all edges between these nodes in $\mathcal{G}$ turned into undirected edges in this new graph, and with new undirected edges connecting each pair of nodes in $\text{bd}(\mathcal{N}_i)$ that were not joined already; then each $\mathbb{P}(\mathcal{N}_i \mid \text{bd}(\mathcal{N}_i))$ equals the ratio $h_i(\mathcal{N}_i, \text{bd}(\mathcal{N}_i)) / (\sum h_i(\mathcal{N}_i, \text{bd}(\mathcal{N}_i)))$, with the sum extending over all configurations of $\mathcal{N}_i$, for appropriate positive functions h_i that factorize according to the corresponding undirected graphs.

3. Logical Credal Networks

A Logical Credal Network (LCN) consists of a set of propositions $\mathcal{N}$ and two sets of constraints $\mathcal{T}_U$ and $\mathcal{T}_D$. The set $\mathcal{N}$ is finite with propositions $A_1, \dots, A_n$. Each proposition A_i is associated with a random variable X_i that is an indicator variable: if A_i holds in an interpretation of the propositions then $X_i = 1$; otherwise, $X_i = 0$. From now on we simply use the same symbol for a proposition and its corresponding indicator random variable. Each constraint in $\mathcal{T}_U$ and in $\mathcal{T}_D$ is of the form

$$\alpha \leq \mathbb{P}(\phi \mid \varphi) \leq \beta,$$

where each ϕ and each φ is a formula. Note that there is no syntactic difference between formulas in $\mathcal{T}_U$ and in $\mathcal{T}_D$; the difference between these sets lies in the distinct ways their formulas are interpreted (as discussed later).

In this paper, formulas are propositional (with propositions in $\mathcal{N}$ and connectives such as negation, disjunction, conjunction). The definition of LCNs by Marinescu et al. [22] allows for relational structures and first-order formulas; however, their semantics is obtained by grounding on finite domains, so we can focus on a propositional language for our purposes here.

Note that φ can be a tautology $\top$, in which case we can just write the “unconditional” probability $\mathbb{P}(\phi)$. One can obviously use simple variants of constraints, such as $\mathbb{P}(\phi \mid \varphi) = \beta$ or $\mathbb{P}(\phi \mid \varphi) \geq \alpha$ or $\mathbb{P}(\phi) \leq \alpha$, as syntactic sugar whenever needed. We use that in our examples.

The semantics of an LCN is given by a translation from the LCN to a directed graph, referred to as the *primal graph* of the LCN. In the primal graph, each proposition in the LCN is a node (we often refer to such nodes as *proposition-nodes*). Each constraint of the form $\alpha \leq \mathbb{P}(\phi \mid \varphi) \leq \beta$ is processed as follows.

1. First, a node labeled with formula ϕ is added, and an edge is added from node ϕ to each proposition in φ .⁴

⁴We note that the original presentation of LCNs [22] is a bit different from the one adopted here, as no edges are added for a constraint in $\mathcal{T}_D$ for which φ is $\top$. But this does not make any difference in the results and simplifies the presentation. In fact, a more recent description of LCNs [23] adopts the same scheme.

2. In case the constraint is in $\mathcal{T}_U$, an edge is added from each proposition in ϕ to node ϕ .
3. In case φ is not $\top$, a node labeled with φ is added, a directed edge from φ to ϕ is added, and an edge from each proposition in φ to node φ is added.

We often refer to nodes labeled with formulas as *formula-nodes*. Note the difference between formulas in $\mathcal{T}_U$ and in $\mathcal{T}_D$: a formula in the former set leads to bi-directed edges between the node representing the formula and its “conditioned” propositions (that is, propositions in ϕ), while a formula in the latter set only leads to directed edges from the node representing the formula to its “conditioned” atoms. As we later introduce and study *structures* of LCNs, it will be clear that a formula in $\mathcal{T}_U$ introduces undirected edges among all propositions in ϕ , thus eliminating the possibility that these propositions are independent, while a formula in $\mathcal{T}_D$ leaves open the possibility that the propositions are independent (they may become dependent through other paths imposed by other formulas). This distinction was proposed so as to enhance the level of expressivity at a small cost in language complexity.

As shown in the next example, in our drawings formulas appear inside dashed rectangles. For the sake of simplicity, we remove such a rectangle whenever the corresponding formula contains a single proposition and its connections to surrounding nodes can be inferred from context; in such cases we can just connect edges from and to the corresponding proposition-nodes.

Example 1. Consider the following LCN, based on the Smokers and Friends example by Marinescu et al. [22]. We have propositions C_i, F_i, S_i for $i \in \{1, 2, 3\}$. All constraints belong to $\mathcal{T}_U$ (that is, $\mathcal{T}_D$ is empty), with $i, j \in \{1, 2, 3\}$:

$$\begin{aligned} 0.5 &\leq \mathbb{P}(F_i | F_j \wedge F_k) \leq 1, & i \neq j, i \neq k, j \neq k; \\ 0 &\leq \mathbb{P}(S_i \vee S_j | F_i) \leq 0.2, & i \neq j; \\ 0.03 &\leq \mathbb{P}(C_i | S_i) \leq 0.04; \\ 0 &\leq \mathbb{P}(C_i | \neg S_i) \leq 0.01. \end{aligned}$$

The primal graph of this LCN is depicted in Figure 3. Note that there are several directed cycles in this primal graph. $\square$

Marinescu et al. [22] then define:

Definition 4. An *lcn-parent* of a proposition A is a proposition such that there is a directed path in the primal graph from it to A in which all intermediate nodes are formulas.

The set of lcn-parents of A is denoted by $\text{lcn-pa}(A)$.

Definition 5. An *lcn-descendant* of a proposition A is a proposition such that there is a directed path in the primal graph from A to it in which no intermediate node is an lcn-parent of A .

The set of lcn-descendants of A is denoted by $\text{lcn-de}(A)$.

The connections between these concepts and the definitions of parent and descendant in Section 2 are studied and clarified in the next section.

Using these definitions, Marinescu et al. [22] proposed a Markov condition:

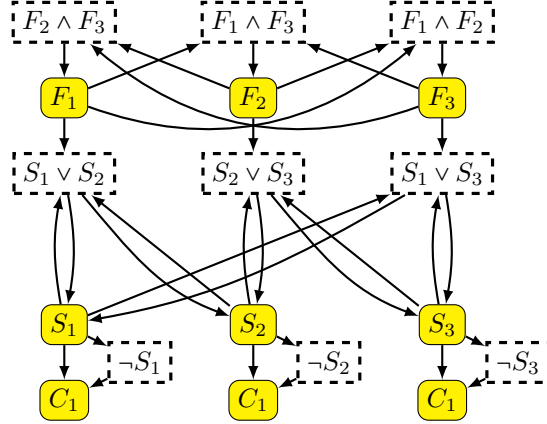


Figure 3: The primal graph of the LCN in Example 1.

Definition 6 (LMC(LCN)). A node A is independent, given its lcn-parents, of all nodes that are not A itself nor lcn-descendants of A nor lcn-parents of A .

The Markov condition in Definition 6 can be expressed as follows, using $\perp\!\!\!\perp$ to mean “is independent of”:

$$A \perp\!\!\!\perp \mathcal{N} \setminus \{\{A\} \cup \text{lcn-de}(A) \cup \text{lcn-pa}(A)\} \mid \text{lcn-pa}(A), \quad (2)$$

We use $\perp\!\!\!\perp$ similarly in the remainder of the paper. We also use the superscript c to mean complement, hence $\mathcal{A}^c \doteq \mathcal{N} \setminus \mathcal{A}$. That is, the Markov condition in Definition 6 can be written

$$A \perp\!\!\!\perp \{\{A\} \cup \text{lcn-de}(A) \cup \text{lcn-pa}(A)\}^c \mid \text{lcn-pa}(A).$$

Marinescu et al. [22] have derived inference algorithms (that is, they consider the computation of conditional probabilities) that exploit such independence relations, and they have looked at applications that demonstrate the practical value of LCNs.

It seems that a detailed discussion about the meaning of this Markov condition, as well as its properties and consequences, would be welcome. To do so, we find it useful to introduce a novel concept, namely, the *structure* of an LCN.

4. The Structure of an LCN

The primal graph of an LCN is rather similar in spirit to the *factor graph* of a Bayesian network [19], where both random variables and conditional probabilities are explicitly represented. This is a convenient device when it comes to message-passing inference algorithms [19, 24], and indeed it has been used to develop message-passing schemes [23]; however, perhaps it contains too much information when one wishes to examine independence relations.

We introduce another graph, to be extracted from the primal graph of a given LCN, as follows:

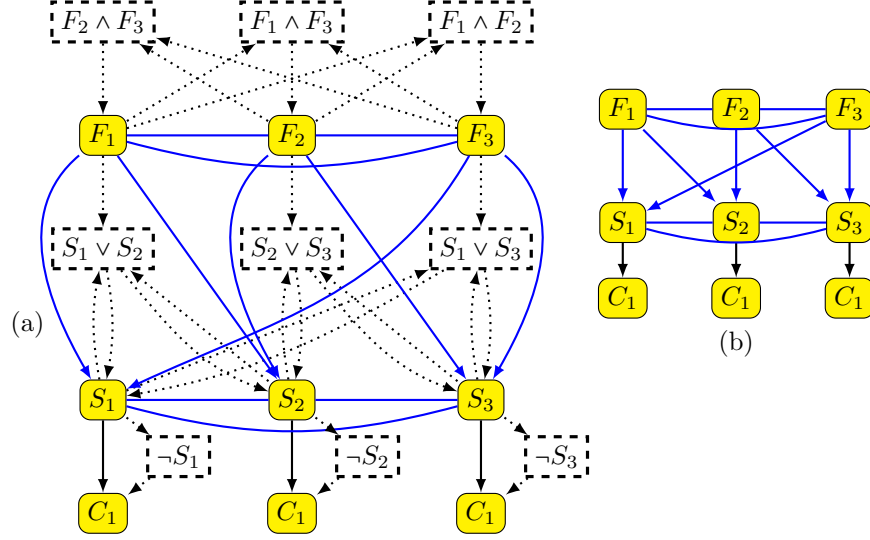


Figure 4: (a) The primal graph for the LCN in Example 1, together with the edges in the structure of the LCN. Edges in the structure are solid (the ones added in the process are in blue); edges in and out of formula-nodes are dotted. (b) The structure of the LCN, by removing the formula-nodes and associated edges.

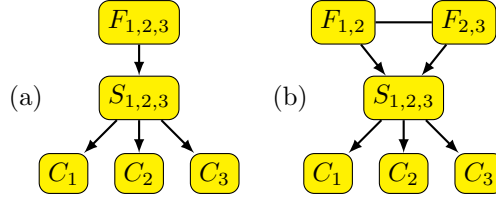


Figure 5: (a) A directed acyclic graph with the chain components of the chain graph that represents the structure. (b) A variant discussed in Example 3.

1. For each formula-node ϕ that appears as a conditioned formula in a constraint in $\mathcal{T}_U$, place an undirected edge between any two propositions that appear in ϕ .
2. For each pair of formula-nodes φ and ϕ that appear in a constraint, add a directed edge from each proposition in φ to each proposition in ϕ .
3. If, for some pair of proposition-nodes A and B , there is now a pair of edges $A \rightarrow B$, then replace both edges by an undirected edge.
4. Finally, remove the formula-nodes and all edges in and out of them; and if there are identical edges left, remove duplicates.

This graph is referred to as the *structure* of the LCN.

Example 2. Figures 4 depicts the structure of the LCN in Example 1, both superimposed on the primal graph (left) and by itself (right). $\square$

We have:

Lemma 1. *The set of lcn-parents of a proposition A in an LCN is identical to the boundary of A with respect to the structure of the LCN.*

Proof. Consider an LCN with a primal graph $\mathcal{D}$. If B is an lcn-parent of A with respect to $\mathcal{D}$, then constructs such as $B \rightarrow \varphi \rightarrow \phi \rightarrow A$ or $B \rightarrow \phi \rightarrow A$ must be present, leading either to directed or undirected edges in the structure; hence B appears either as a parent or a neighbor of A in the structure of the LCN. Conversely, if B is a parent or a neighbor of A in the structure of the LCN, then one of the sequences of edges already mentioned must be in $\mathcal{D}$, so B is an lcn-parent of A in $\mathcal{D}$. $\square$

The most natural candidate for the concept of “descendant” in a structure, so as to mirror the concept of lcn-descendant, seems to be:

Definition 7. If there is a directed path from A to B such that no intermediate node is a boundary node of A , then B is a *strict descendant* of A .

The set of strict descendants of A is denoted by $\text{sde}(A)$.

However, the lcn-descendants of a node in a primal graph are not the strict descendants of the same node in the corresponding structure. To see this, consider Figure 1.b: if we think of this graph as containing the proposition nodes and their connections from the primal graph, then A and D are the lcn-descendants of B ; in the corresponding structure, given by Figure 1.c, B has no strict descendants as there are no directed paths.

In any case, we can state a local Markov condition that we can show to have the same consequences as LMC(LCN):

Definition 8 (LMC(C-STR)). A node A is independent, given its boundary, of all nodes that are not A itself nor strict descendants of A nor boundary nodes of A .

In symbols, the LMC(C-STR) imposes

$$A \perp\!\!\!\perp \mathcal{N} \setminus \{\{A\} \cup \text{sde}(A) \cup \text{bd}(A)\} \mid \text{bd}(A). \quad (3)$$

It is not immediately clear that LMC(LCN) and LMC(C-STR) are equivalent when applied to a structure, given the difference between lcn-descendants and strict descendants. So we must examine these concepts in more detail.

The set of lcn-descendants of a proposition node A , $\text{lcn-de}(A)$, can be divided in two sets of nodes, $\text{lcn-de}_1(A)$ and $\text{lcn-de}_2(A)$. Take a node B to be in $\text{lcn-de}_1(A)$ if and only if there is at least one directed path from A to B that starts with a directed edge and such that no intermediate node is an lcn-parent of A . All of those nodes are strict descendants of A because the directed paths that exist in the primal graph are translated verbatim to directed paths in the corresponding structure. And take a node B to be in $\text{lcn-de}_2(A)$ if and only if all directed paths from A to B that have no lcn-parent of A as an intermediate node start with a bi-directed edge. By construction, $\text{lcn-de}_1(A)$ and $\text{lcn-de}_2(A)$ are disjoint. Now note that any node B in $\text{lcn-de}_2(A)$ must be an lcn-parent of A (if there is B that is not an lcn-parent of A , we have a situation where all paths from A to B go through an lcn-parent of A , and by definition this is not possible). Note also that nodes in $\text{lcn-de}_2(A)$ appear as neighbors of A in the corresponding structure of the LCN.

Consequently, the nodes that are lcn-descendants of A are *either* strict descendants of A *or* neighbors of A with respect to the structure.

Using these insights, we can show that the LMC(LCN) and the LMC(C-STR) do have the same effect when applied to the structure:

Theorem 1. *Given an LCN, the Markov condition LMC(LCN) in Definition 6 is identical to the local Markov condition LMC(C-STR) in Definition 8 with respect to the independence relations the first condition imposes, when applied to the structure of the LCN.*

Proof. To prove that Expressions (2) and (3) are equivalent, consider first the following equalities with respect to sets of nodes:

$$\begin{aligned} \text{lcn-de}(A) \cup \text{lcn-pa}(A) &= \text{lcn-de}_1(A) \cup \text{lcn-de}_2(A) \cup \text{lcn-pa}(A) \\ &= \text{sde}(A) \cup \text{ne}(A) \cup \text{pa}(A) \cup \text{ne}(A) \\ &= \text{sde}(A) \cup \text{pa}(A) \cup \text{ne}(A) \\ &= \text{sde}(A) \cup \text{bd}(A), \end{aligned}$$

and consequently $\{A\} \cup \text{lcn-de}(A) \cup \text{lcn-pa}(A)$ is equal to $\{A\} \cup \text{sde}(A) \cup \text{bd}(A)$, hence

$$(\{A\} \cup \text{lcn-de}(A) \cup \text{lcn-pa}(A))^c = (\{A\} \cup \text{sde}(A) \cup \text{bd}(A))^c.$$

Now note that $\text{lcn-pa}(A) = \text{bd}(A)$, by Lemma 1, to obtain the desired result. $\square$

In the presence of directed cycles in a primal graph, we may have lcn-parents that are lcn-descendants. Similarly, in a structure we may have parents and

neighbors that are strict descendants. The only real difference between lcn-descendants and strict descendants surfaces in a particular case: for a given node A , there may be a node B that is a neighbor of A in the structure and that is not reached through any directed cycle emanating from A ; this node B is an lcn-descendant of A in the originating primal graph, but not a strict descendant of A with respect to the structure. For instance, if Figure 1.b is viewed as the primal graph of an LCN, then Figure 1.c carries its structure; note that B is an lcn-descendant of A in the primal graph but not a strict descendant of A in the corresponding structure. This difference however does not lead to a difference between $\text{LMC}(\text{LCN})$ and $\text{LMC}(\text{C-STR})$, as shown by Theorem 1.

5. Chain Graphs and Factorization

If the structure of an LCN is a directed acyclic graph, the $\text{LMC}(\text{LCN})$ mimics the usual local Markov conditions for Bayesian networks (also adopted in credal networks [8, 25]). If instead all constraints in an LCN belong to $\mathcal{T}_U$ and all of them only refer to “unconditional” probabilities (that is, $\varphi = \top$ in every constraint), then the structure of the LCN is an undirected graph endowed with the usual local Markov condition for undirected graphs.

These two points can be generalized in a satisfying way *whenever the structure contains no directed cycle*:

Theorem 2. *If the structure of an LCN is a chain graph, and probabilities are positive, then the Markov condition $\text{LMC}(\text{LCN})$ in Definition 6 is identical, with respect to the independence relations it imposes, to the $\text{LMC}(C)$ applied to the structure.*

Before we prove this theorem, it should be noted that sets of descendants and of strict descendants are not identical. This is easy to see in graphs with directed cycles: in Figure 1.b, node B has descendants $\{A, C, D\}$ and strict descendants $\{A, D\}$. But even in chain graphs we may have differences: for instance, suppose that in Figure 1.e we add a single directed edge from D to a new node E ; then E is the only descendant of B , but B has no strict descendants.

In fact, the descendants of a node A can be divided into two sets, much as we did in connection with the lcn-descendants of a node. A node B is in the first set if and only if there is at least one directed path from A to B that starts with a directed edge. All of those nodes are strict descendants of A . A node C is in the second set if and only if all directed paths from A to C start with an undirected edge. Then all directed paths from A to C first reach a node that is in the boundary of A and consequently C is *not* a strict descendant of A . The first set is thus exactly $\text{sde}(A)$, and we might refer to the second set as $\text{wde}(A)$, the set of “weak” descendants of A . By construction we have $\text{sde}(A) \cap \text{wde}(A) = \emptyset$. Moreover, we conclude (see Figure 6) that

$$(\text{sde}(A))^c = (\text{de}(A))^c \cup \text{wde}(A),$$

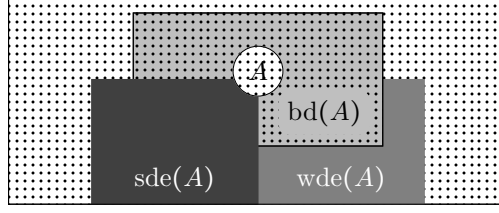


Figure 6: Proposition nodes of an LCN whose structure is a chain graph. The dotted area contains the nodes that are not descendants. In a chain graph, the strict descendants cannot be in the boundary.

from which we have that

$$\mathcal{N} \setminus (\{A\} \cup \text{sde}(A) \cup \text{bd}(A)) = \{A\}^c \cap ((\text{de}(A))^c \cup \text{wde}(A)) \cap (\text{bd}(A))^c.$$

As $(\{A\}^c \cap \text{wde}(A) \cap (\text{bd}(A))^c) = \text{wde}(A)$ by construction, the set

$$\mathcal{N} \setminus (\{A\} \cup \text{sde}(A) \cup \text{bd}(A))$$

is the union of two disjoint sets:

$$(\{A\}^c \cap (\text{de}(A))^c \cap (\text{bd}(A))^c) \cup \text{wde}(A). \quad (4)$$

We can now prove Theorem 2.

Proof. Assume the LMC(C-STR); so, for any node A , we have Expression (3). Using Expression (4), we obtain

$$A \perp\!\!\!\perp ((\{A\}^c \cap (\text{de}(A))^c \cap (\text{bd}(A))^c) \cup \text{wde}(A)) \mid \text{bd}(A).$$

Using the Decomposition property of probabilities,⁵ we reach $A \perp\!\!\!\perp \{A\}^c \cap (\text{de}(A))^c \cap (\text{bd}(A))^c \mid \text{bd}(A)$. Thus the LMC(C) holds.

Assume the LMC(C) holds; given the positivity condition, the GMC(C) holds [7]. Take a node A and suppose there is a node $B \in \mathcal{N} \setminus (\{A\} \cup \text{sde}(A) \cup \text{bd}(A))$. We now prove that B is separated from A by $\text{bd}(A)$ in

$$\begin{aligned} \mathcal{G}_A^* &\doteq \mathcal{G}^{ma}(\{A\} \cup (\{A\}^c \cap (\text{sde}(A))^c \cap (\text{bd}(A))^c) \cup \text{bd}(A)) \\ &= \mathcal{G}^{ma}((\text{sde}(A))^c), \end{aligned}$$

and consequently the GMC(C) leads to

$$A \perp\!\!\!\perp (\{A\}^c \cap (\text{sde}(A))^c \cap (\text{bd}(A))^c) \mid \text{bd}(A)$$

as desired. Note that, by construction, nodes in $\text{sde}(A)$ cannot be in the ancestral set of any node in $\text{wde}(A)$, so $\text{sde}(A)$ is not in $\mathcal{G}_A^*$. Consequently, $\mathcal{G}_A^*$ is the

⁵The Decomposition property states that $X \perp\!\!\!\perp Y \cup W \mid Z$ implies $X \perp\!\!\!\perp Y \mid Z$ for sets of random variables W, X, Y, Z [19].

moral graph of the graph consisting of nodes in $(\text{sde}(A))^c$ and edges among them in the structure. In the moral graph, paths from A to B may start by moving from A to a parent or a neighbor of A in the original structure of the LCN, and those paths are blocked by $\text{bd}(A)$. The only other way to reach B from A in $\mathcal{G}_A^*$ would be to follow a path $\mathcal{G}_A^*$ starting with an edge connecting A and some node C such that A and C are both parents of a common chain component; however this cannot happen as all direct children of A are in $\text{sde}(A)$, and $\mathcal{G}_A^*$ does not contain nodes in $\text{sde}(A)$. Hence we must have separation of A and B by $\text{bd}(A)$ in $\mathcal{G}_A^*$, and therefore we have separation of A and $\mathcal{N} \setminus (\{A\} \cup \text{sde}(A) \cup \text{bd}(A))$ by $\text{bd}(A)$ as desired. $\square$

The significance of the previous theorem is that, assuming that all probabilities are positive, the local Markov condition for a chain graph is equivalent both to the global Markov condition and to the factorization property of chain graphs. A factorization allows us to break down the probability distribution over all random variables in an LCN in hopefully much smaller pieces that require less specification effort.

Example 3. Figure 4.b depicts a structure that is in fact a chain graph. We can group variables to obtain chain components $F_{1,2,3}$ and $S_{1,2,3}$ and draw a directed acyclic graph with the chain components, as in Figure 5.a. The joint probability distribution factorizes as Expression (1):

$$\begin{aligned} \mathbb{P}(F_{1,2,3} = f, S_{1,2,3} = s, C_1 = c_1, C_2 = c_2, C_3 = c_3) &= \\ \mathbb{P}(F_{1,2,3} = f) \mathbb{P}(S_{1,2,3} = s | F_{1,2,3} = f) & \\ \mathbb{P}(C_1 = c_1 | S_{1,2,3} = s) \mathbb{P}(C_2 = c_2 | S_{1,2,3} = s) \mathbb{P}(C_3 = c_3 | S_{1,2,3} = s), & \end{aligned}$$

where f is a configuration of the random variables in $F_{1,2,3}$, while s is a configuration of the random variables in $S_{1,2,3}$. Because there are no independence relations “inside” the chain components, this factorization is guaranteed even if some probability values are equal to zero [7].

Suppose that the three constraints $0.5 \leq \mathbb{P}(F_i | F_j \wedge F_k) \leq 1$ are replaced so that, instead of a single chain component with F_1, F_2, F_3 , we have two chain components, one with F_1 and F_2 , the other with F_2 and F_3 . The chain components might be organized as in Figure 5.b. If all probabilities are positive, that chain graph leads to a factorization of the joint probability distribution similar to the previous one, but now

$$\mathbb{P}(F_{1,2,3} = f_1 f_2 f_3) = \phi_1(F_{1,2} = f_1 f_2) \phi_2(F_{2,3} = f_2 f_3),$$

where ϕ_1 and ϕ_2 are positive functions, and the values of F_1, F_2 and F_3 are indicated by f_1, f_2, f_3 respectively. $\square$

The assumption that probabilities are positive is important: when some probabilities are zero, there is no guarantee that a factorization actually exists [26]. This is unfortunate as a factorization leads to valuable computational simplifications. One strategy then is to guarantee that all configurations do

have positive probability, possibly by adding language directives that bound probabilities from below. This solution may be inconvenient if we do have some hard constraints in the domain. For instance, one may wish to impose that $A \vee B$ holds (in which case $\mathbb{P}(\neg A \wedge \neg B) = 0$).

However, it is still possible to obtain a factorization even when hard constraints are imposed. Say we have a formula, for instance $A \vee B$, that must be satisfied. We treat it as a constraint $1 \leq \mathbb{P}(A \vee B) \leq 1$ in $\mathcal{T}_U$, thus guaranteeing that there is a clique containing its propositions/random variables. Then we remove the impossible configurations of these random variables (in our running example, we remove $A = B = 0$), thus reducing the number of possible configurations for the corresponding clique. A factorization is obtained again in the reduced space of configurations, provided the remaining configurations do have positive probabilities.

Finally, an entirely different strategy may be pursued: adopt a stronger Markov condition that guarantees factorization (and hence global independence relations) in all circumstances. Moussouris [26] has identified one such condition, where a system is *strongly Markovian* in case a Markov condition holds for the system and suitable sub-systems. That (very!) strong condition forces zero probabilities to be, in a sense, localized, so that probabilities satisfy a nice factorization property. Alas, the condition cannot be guaranteed for all graphs, and its consequences have not been explored in depth so far.

6. Directed Cycles, and Alternative Semantics

As noted already, existence of a factorization is a very desirable property for any probabilistic formalism: not only it simplifies calculations, but it also emphasizes modularity in modeling and ease of understanding. In the previous section we have shown that LCNs whose structure is a chain graph do have, under a positivity assumption, a well-known factorization property. We now examine how that result may be extended to more general settings.

Clearly, the LMC(LCN) is a local condition that can be applied even in the presence of directed cycles — there are no technical problems with it. However, a local Markov condition may not be very satisfactory in the presence of directed cycles, as a simple yet key example suggests:

Example 4. Take an LCN whose primal graph is a long cycle, as depicted in Figure 7. No A_i has any non-descendant non-parent. And no A_i has any non-strict-descendant non-parent distinct from A_i . The local Markov conditions we have contemplated do not impose *any* independence relation. $\square$

Local conditions seem rather weak when there are long directed cycles. On the other hand, a global condition does produce sensible independence relations within a long cycle. For instance, the GMC(C) in Example 4 produces non-trivial independence relations such as $A_2 \perp\!\!\!\perp A_4 | A_1, A_3$ (assuming $k > 3$). Indeed, in that example we have $A_1 \perp\!\!\!\perp A_3, \dots, A_{k-1} | A_2, A_k$, $A_2 \perp\!\!\!\perp A_4, \dots, A_k | A_1, A_3$, and,

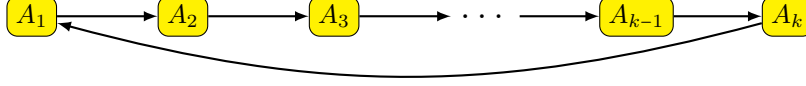


Figure 7: A primal graph with a very long cycle.

more generally, for any A_i with $2 < i < k - 2$, we have the independence relation $A_i \perp\!\!\!\perp A_1, \dots, A_{i-2}, A_{i+2}, A_k | A_{i-1}, A_{i+1}$.

Spirites, in seminal work [34], compared the GMC(C) and the LMC(C) in the context of directed graphs that may have cycles. In that case the LMC(C) reduces to the LMC(D). Spirites sided with the GMC(C) so as to obtain a factorization that will be discussed shortly.

We might thus consider the GMC(C) as the basic semantics for LCNs. Suppose then that we start with the primal graph of an LCN of interest, and we then build the following directed graph, a variant of the structure where we keep bi-directed edges:

1. For each formula-node ϕ that appears as a conditioned formula in a constraint in $\mathcal{T}_U$, place a bi-directed edge between any two propositions that appear in ϕ .
2. For each pair of formula-nodes φ and ϕ that appear in a constraint, add a directed edge from each proposition in φ to each proposition in ϕ .
3. Finally, remove the formula-nodes and all edges in and out of them; and if there are identical edges left, remove duplicates.

We refer to the resulting graph as the *full-structure* of the LCN; we might then apply the GMC(C) to such a graph.

The GMC(C) applied to the full-structure may not be equivalent to the LMC(LCN), *even* when the structure is a chain graph (a situation where the GMC(C) applied to the structure yields the same results as the LMC(LCN)).

Example 5. Suppose we have an LCN whose primal graph is depicted in Figure 1.d. To simplify matters suppose all formulas are just individual propositions; there are no formulas with more than one proposition. For instance, we might simply have the assessments $0.1 \leq \mathbb{P}(X|Y) \leq 0.9$ whenever $Y \rightarrow X$ is an edge in Figure 1.d. Take that all configurations are assumed to have positive probability.

While the structure of this LCN is the chain graph in Figure 1.e, its full-structure is identical to the primal graph. The GMC(C) applied to the structure yields $A \perp\!\!\!\perp C$, $B \perp\!\!\!\perp C | A, D$ and $A \perp\!\!\!\perp D | B, C$, the same as the LMC(C) applied to the structure and the LMC(LCN) applied to the primal graph. The GMC(C) applied to the full-structure yields $A \perp\!\!\!\perp C$ and $A \perp\!\!\!\perp C | B, D$, the former because the graph $\mathcal{G}^{ma}(\{A, C\})$ contains just two disjoint nodes A and C , and the latter because the graph $\mathcal{G}^{ma}(\{A, B, C, D\})$ is depicted in Figure 2.d. $\square$

Significant recent results can help build additional insight about the behavior of these various Markov conditions, even if no complete theory can be assembled

for general LCNs at this time. In particular, Forré and Mooij have captured a significant fraction of the relevant literature in a pair of long manuscripts [12, 13], from key early efforts by Spirtes [34], Richardson [31], Koster [20] and others, to results connected with structural causal models [6, 17].

The first positive aspect of the GMC(C), when applied to the full-structure, is the existence of a factorization, already alluded in connection with the work of Spirtes [34]. The result requires one more concept: for a directed graph, a *strong connected component* (not to be confused with the chain components defined previously) is a set of nodes such that it is the intersection of ancestors and descendants of each one of its nodes [12, Definition 2.1.2]. Intuitively, the strong connected components are the “largest possible cycles” that can be built by freely visiting nodes. Now, assuming all probabilities are positive, the GMC(C) is equivalent to a factorization such that $\mathbb{P}(X = x) = \prod_i \prod_{j_i} \phi_{ij}(x_{ij})$, where the index i runs over strong connected components; the index j_i runs over the cliques of the moral graph of the smallest ancestral set of the i strong connected components; and ϕ_{ij} is a positive function over the elements of the corresponding clique [12, Corollary 3.6.9, Lemma 3.6.14].

Another aspect of the GMC(C) that has received attention is the fact that the GMC(C) may fail in relatively simple circumstances. For instance, a set of structural equations on binary variables may have a single satisfying joint distribution that agrees with the LMC(D) but that fails the GMC(C) [27]. There is now a rather complete understanding about conditions that prevent such inconsistencies; in particular, σ -separation has been proposed as a basis for global Markov conditions so as to guarantee consistency with structural equations [12]. A promising path for future investigation is to study the consequences of σ -separation as a semantics for LCNs.

In short, if we keep the LMC(LCN), it seems difficult to find a condition that yields a factorization; on the other hand, if we impose some version of the GMC(C) to primal graphs to obtain a factorization, it seems difficult to stay with the LMC(LCN). Further analysis of practical problems seems necessary to decide which path to follow.

The remainder of this section takes a more speculative tone as we contemplate variants of the original LCN semantics, by applying the GMC(C) to structures and related graphs. This investigation is motivated by the weakness of local conditions as shown by Example 8, and also by the fact that a condition based on moralized graphs is the most obvious route to factorization properties (as the Hammersley-Clifford theorem can then be invoked under a positivity assumption [26]).

Here is a possible semantics that simply formalizes the sort of strategy we have touched upon in previous paragraphs: take an LCN to represent the set of probability distributions over its nodes such that all constraints in the LCN are satisfied, and each distribution satisfies the GMC(C) *with respect to the structure*. As the GMC(C) is equivalent to the LMC(LCN) when a structure is a chain graph, we do preserve the LMC(LCN) in those cases. However, recall that the GMC(C) and the LMC(C) may differ in the presence of directed cycles (Example 4). Factorization can once more rely on previous results in the

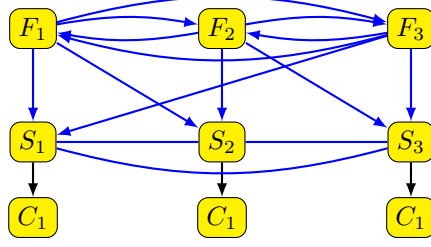


Figure 8: The mixed-structure for the LCN in Example 1.

literature, at least for important cases: Koster [20, Theorem 3.4] has already proven factorization properties for *reciprocal graphs*, a large class of graphs that allow for directed and undirected edges and that allow for cycles. Enlarging those results seems a relevant avenue for future work.

When we think of structures as the primary tool to determine independence relations, we might wish to enhance structures so as to differentiate connections that arise in distinct ways. A possible strategy would then be as follows. Take an LCN and build a *mixed-structure* by going through following steps:

1. For each constraint $\alpha \leq \mathbb{P}(\phi|\varphi) \leq \beta$ in $\mathcal{T}_U$, add an undirected arrow between each pair of proposition-nodes in ϕ .
2. For each constraint $\alpha \leq \mathbb{P}(\phi|\varphi) \leq \beta$ add a directed edge from each proposition-node in φ to each proposition-node in ϕ (if φ is $\top$, there is no such edge to add).
3. Remove multiple identical edges.

That is, create a node per proposition that appears in the LCN; then take each constraint in $\mathcal{T}_U$ and add undirected edges between any two propositions in ϕ , and finally take each constraint and add a directed edge from each proposition that appears in φ to each proposition that appears in the corresponding ϕ . Figure 8 depicts the mixed-structure for Example 1.

Therefore, we have that an LCN represents the set of probability distributions over its nodes such that all constraints in the LCN are satisfied, and each distribution satisfies the $\text{GMC}(\mathcal{C})$ with respect to the mixed-structure.

The next example emphasizes the differences between semantics.

Example 6. Suppose we have an LCN with constraints $\mathbb{P}(B|A) = 0.2$, $\mathbb{P}(D|E) = 0.3$, $\mathbb{P}(B \vee C) = 0.4$, $\mathbb{P}(C \vee D) = 0.5$. Both the structure and the mixed-structure of this LCN is depicted in Figure 9.a. Consider another LCN with constraints $\mathbb{P}(B|A \wedge C) = 0.2$, $\mathbb{P}(C|B \wedge D) = 0.3$, and $\mathbb{P}(D|C \wedge E) = 0.4$. This second LCN has the same structure as the first one, but the mixed-structure are depicted in Figure 9.b. The $\text{GMC}(\mathcal{C})$ produces quite different sets of independence relations when applied to these distinct mixed-structures; for instance, $A, B \perp\!\!\!\perp D|C, E$ in the first LCN, but not necessarily in the second; $A, B \perp\!\!\!\perp E|C, D$ in the second LCN, but not necessarily in the first. This seems appropriate as the LCNs con-



Figure 9: Structures and mixed-structures in Example 6.

vey quite distinct scenarios, one related to the symmetry of logical constraints, the other related to the links induced by directed influences. $\square$

We hope to pursue a comparison between the theoretical and pragmatic aspects of these semantics in future work. It is worth noticing that there are many other relevant proposals that deserve further attention as possible variants on the original LCN syntax and semantics. For instance, *dependency networks* [16] allow for directed cycles and do have a modular specification scheme; they have only an approximate factorization, but that may be enough in applications. Another proposal has been advanced by Schmidt and Murphy [33], where directed cycles are allowed and the adopted Markov condition looks only at the *Markov blanket* of nodes; it does not seem that a factorization has been proven for that proposal, but it is attractive in its simplicity. And a variety of connections with feedback loops and causal mechanisms has been pursued in several applications [4]. There is indeed a huge literature, filled with independence conditions and factorization properties, to which we cannot do justice in the available space. It is necessary to examine whether we can bring elements of those previous efforts into LCNs.

7. Causal Reasoning with LCNs

The discussion has been, so far, rather abstract, and one may wonder how the examined modeling features and Markov conditions fare in practice. Thus we devote this section to a setting where the modeling power of LCNs seems appropriate, as probability intervals arise naturally, and where the effect of distinct Markov conditions may be compared; namely, we look into causal reasoning. Indeed, interventional and counterfactual queries [28] often require probability bounds and may involve cyclic feedback connections. Thus it seems natural to examine such queries when analyzing the power of LCNs, even more so when one notes that causal reasoning must be an essential feature of automated agents.

The key issue we intend to explore in this section is failure of *identifiability* for some parameters, and the consequences of such failures on interventional and counterfactual queries, in particular in cases where partial identifiability requires one to handle probability bounds even in the presence of infinitely large datasets [21].

We examine examples that contain both structural causal models and causal diagrams. With respect to structural causal models, we follow the analysis by Zaffalon et al. [36] so as to connect partial identifiability with computation of LCN inferences (that is, of probability bounds for an LCN). Zaffalon et al.

[36] have shown that acyclic structural causal models can be viewed as credal networks — hence, such models can be easily translated to LCNs so as to handle interventional and counterfactual queries. We discuss this translation here and move to the more complex situation where a structural causal model or a causal diagram has cycles, as those cases are not captured by previous literature. We also comment on the computations demanded by Pearl’s theory of counterfactual reasoning, where probability bounds are also often needed. We avoid a detailed discussion of causal reasoning, a complex and vast topic that has been examined in recent textbooks [18, 28].

We start with settings where probabilities related to observable quantities are precisely known, possibly due to the fact that they were estimated using very large amounts of data. For the sake of simplicity, as often done in causal reasoning, we assume (unrealistically) that the number of unobserved random variables, and their values, are known. Our goal there is to show that, *even in the limit of infinitely many observations or precise specification of probabilities*, causal reasoning calls for a proper representation for probability intervals. We note that there may be other sources of imprecision in probability values; one may have an incomplete understanding of substantive assumptions in a domain, or one may have to aggregate disparate opinions of several experts [35]; we return to this comment in our last example, where we put together feedback cycles and imprecise assessments, a combination that has not been examined so far in the literature. There may thus be scenarios where observational data and background assumptions may be captured by formalisms from probability intervals to p-boxes or belief functions [2].

We start with a brief review of structural causal models, to set up the stage. We closely follow the presentation by Zaffalon et al. [36]. We then examine examples to display the expressive power of LCNs and their associated inference algorithms.

A *structural equation*, written

$$X = f_X(Y_1, \dots, Y_m),$$

uses a function f_X to determine the value of variable X from the values of $Y_1, \dots, Y_m$, $m > 0$. Then $Y_1, \dots, Y_m$ are referred to as *parents* of X , denoted by $\text{pa}(X)$. A *structural causal model* (SCM) consists of two sets of variables and a set of structural equations: We have a set $\mathbf{X}$ of random variables, such that for each $X \in \mathbf{X}$ there is a corresponding structural equation whose function f_X may depend on variables in $\mathbf{X}$ and also on random variables in a set $\mathbf{U}$. Elements of $\mathbf{X}$ are referred to as *endogenous variables* and elements of $\mathbf{U}$ are referred to as *exogenous variables*. The *endogenous parents* of endogenous variable X are elements of $\text{pa}(X) \cap \mathbf{X}$. The *causal diagram* of an SCM is a directed graph where: (1) the set of nodes is the set of endogenous or exogenous variables, (2) for each endogenous variable X , there are directed edges from the parents of X to X . A *probabilistic SCM* consists of a pair: an SCM and a probability distribution over the exogenous variables of the SCM. If the causal diagram of a probabilistic SCM has no directed cycle, the SCM is *semi-Markovian*. In a

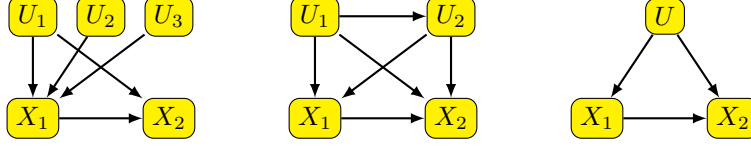


Figure 10: Simple SCMs that represent two binary variables X_1 and X_2 and a confounding mechanism, with independent binary exogenous variables (left), with binary exogenous variables (middle), with a single categorical variable (right).

semi-Markovian SCM, a joint distribution over $\mathbf{U}$ induces a single distribution over $\mathbf{X} \cup \mathbf{U}$.

Our first example offers a tour on how code structural models using LCNs, while introducing a few pieces of syntactic sugar that simplify descriptions. The relatively long, yet simple, example looks at scenarios where the joint distribution over $\mathbf{X}$ is precisely known. That does not imply that a single joint distribution over $\mathbf{X} \cup \mathbf{U}$ is fixed; rather, a set of such distributions over $\mathbf{X} \cup \mathbf{U}$ may be compatible with the given distribution over $\mathbf{X}$. In this example the global and local Markov conditions are identical and not affected by zero probabilities (due to acyclicity).

Example 7. Suppose one has the SCM with structural equations:

$$X_1 = \neg U_1 \wedge U_2 \wedge U_3, \quad X_2 = X_1 \vee U_1. \quad (5)$$

where all variables are binary (values 0 and 1) and we abuse notation by placing them in Boolean expressions (taking 0 to mean false and 1 to mean true). The corresponding causal diagram is depicted in Figure 10 (left). Note that we write $X = f_X(Y_1, \dots, Y_m)$ to indicate that this equality holds with probability one through suitable assessments of X conditional on values of $Y_1, \dots, Y_m$. This is a useful piece of syntactic sugar. Now suppose we gather information about X_1 and X_2 to obtain:

$$\mathbb{P}(\neg X_1 \wedge \neg X_2) = 1 - \alpha - \beta, \quad \mathbb{P}(\neg X_1 \wedge X_2) = \alpha, \quad \mathbb{P}(X_1 \wedge X_2) = \beta, \quad (6)$$

where $\alpha \in (0, 1)$ and $\beta \in (0, 1)$. Now denote $\mathbb{P}(U_i = 1)$ by q_i . While $q_1 = \alpha$, both q_2 and q_3 vary within identical intervals $[k, 1]$ where $k = \beta/(1 - \alpha)$, as one can obtain by running an inference algorithm on the LCN [23]. (The vectors $[q_1, q_2, q_3]$ can in fact be parameterized as $[\alpha, t, k/t]$, with $t \in [k, 1]$.)

A variant, with just two exogenous variables, might be given by:

$$X_1 = U_1 \wedge U_2, \quad X_2 = X_1 \vee (U_1 \wedge \neg U_2) \vee (\neg U_1 \wedge U_2),$$

plus assessments in Expression (6). The corresponding causal diagram is depicted in Figure 10 (middle). Denote $\mathbb{P}(U_1 = i, U_2 = j)$ by q_{ij} . Now the possible vectors $[q_{00}, q_{01}, q_{10}, q_{11}]$ can be parameterized as $[1 - \alpha - \beta, t, \alpha - t, \beta]$, with $t \in [0, \alpha]$.

Instead of introducing several exogenous variables, we may introduce another piece of syntactic sugar, where we allow for categorical exogenous variables and write $\{U = u\}$ to indicate that variable U takes value u . For instance, consider yet another variant example, where we have the following structural equations:

$$X_1 = \{U = u_3\}, \quad X_2 = X_1 \vee \{U = u_1\} \vee \{U = u_2\},$$

and the assessments in Expression (6), where U is an exogenous variables with four values u_0, u_1, u_2, u_3 , while X_1 and X_2 are binary variables. The corresponding causal diagram is depicted in Figure 10 (right). Again, these structural equations and assessments can be coded using an LCN, and probability bounds can be obtained through inference. Denote $\mathbb{P}(U = u_i)$ by q_i ; the possible vectors $[q_0, q_1, q_2, q_3]$ can be parameterized as $[1 - \alpha - \beta, t, \alpha - t, \beta]$, with $t \in [0, \alpha]$. $\square$

Consider now more interesting scenarios where we use inferences with LCNs to deal with interventional and counterfactual reasoning [28]. We need to review some notation. Take a simple intervention where the value of a particular variable X is externally set to x . The key quantity is denoted by $\mathbb{P}(Y = y | \text{do}(X) = x)$, where Y is a set of variables of interest and X is intervened (that is the meaning of the “do” operator) [28]. By definition, the latter probability is obtained by removing the structural equation for X and by imposing value x to all appearances of X in other structure equations.

Is it the case then that an intervention may produce bounds for interventional probabilities over the *observable* (endogenous) variables, even when the observational probability distribution over such variables is precisely known? The next example shows that this can happen even in a simple acyclic setting with a single exogenous variable.

Example 8. Suppose we have three binary variables X_1, X_2, X_3 , each with values 0 and 1. We also have an exogenous categorical variable U with five values $u_0, \dots, u_4$. Consider the following structural equations:

$$\begin{aligned} X_2 &= (X_1 \wedge \{U = u_2\}) \vee \{U = u_3\} \vee \{U = u_4\}, \\ X_3 &= (\neg X_2 \wedge (\{U = u_1\} \vee \{U = u_2\})) \vee (X_2 \wedge \{U = u_4\}). \end{aligned} \quad (7)$$

The corresponding causal diagram is depicted in Figure 11 (left).⁶ We assume that X_1 is an observable variable with a trivial exogenous parent (say U' such that $X_1 = U'$), so we do not add that parent to the model. Suppose we have a precise joint distribution over X_1, X_2, X_3 , with observed assessments

$$\mathbb{P}(X_1 = i, X_2 = j, X_3 = k) = p_{ijk} \quad \text{for all possible } i, j, k; \quad (8)$$

where we assume all p_{ijk} to be positive. Denote $\mathbb{P}(X_2 = j, X_3 = k)$ by p_{jk} (thus $p_{jk} = p_{0jk} + p_{1jk}$) and $\mathbb{P}(X_1 = 1)$ by p (thus $p = \sum_{jk} p_{1jk}$).

⁶A similar causal diagram was employed by Balke and Pearl [5] so as to represent response variables and to compute counterfactual probabilities.

A relatively simple LCN can carry all of those structural equations and assessments, using syntactic conventions previously discussed. Existing algorithms can easily compute probability bounds of interest [23].

Now consider interventions. To be concrete, take the following probability values, borrowed from a similar example by Balke and Pearl [5]:

$$\begin{aligned} p_{000} &= 0.32 \times 0.9, & p_{100} &= 0.02 \times 0.1, \\ p_{001} &= 0.04 \times 0.9, & p_{101} &= 0.67 \times 0.1, \\ p_{010} &= 0.32 \times 0.9, & p_{110} &= 0.17 \times 0.1, \\ p_{011} &= 0.32 \times 0.9, & p_{111} &= 0.14 \times 0.1, \end{aligned} \tag{9}$$

from which we obtain $p = 0.1$, and moreover $p_{00} = 0.29$, $p_{01} = 0.103$, $p_{10} = 0.305$, $p_{11} = 0.302$. Thus we have $\mathbb{P}(X_3 = 1|X_2 = 0) = 0.103/(0.103 + 0.29) = 0.262$ but $\mathbb{P}(X_3 = 1|\text{do}(X_2 = 0)) \in [0.103, 0.114]$. We present a derivation of these bounds in Appendix A.

The question then is how to benefit from the machinery of LCNs to compute such probability bounds arising from interventions. The answer is not immediate, and we offer a solution here.

So, suppose we build an LCN, as with previous examples, using Expressions (7) and (8). We can then create the LCN with “copies” of the endogenous variables that are relevant to an intervention, and have those auxiliary endogenous variables be children of the original exogenous variables. Consider the computation of $\mathbb{P}(X = 1|\text{do}(X_2 = 0))$; to do so, one can compute $\mathbb{P}(X'_3|X'_2 = 0)$ in the enlarged LCN depicted in Figure 12. Note that this causal graph is *exactly* the one obtained when computing counterfactual probabilities in Pearl’s theory (those probabilities will be discussed later). However, here we do not have evidence in the original variables; if we did so, then we would be computing counterfactual probabilities.

As a digression, we note that one may exploit the fact that inference with LCNs requires an optimization program, and so one can minimize/maximize specially crafted expressions that yield the result of interventions. For instance, if one is interested in $\mathbb{P}(X_3 = 1|\text{do}(X_2 = 0))$, it is enough to bound $\mathbb{P}(U = u_1) + \mathbb{P}(U = u_2)$. Or one may be interested in the *causal effect* of an intervention [28]: in this particular example one might be interested in the causal effect of X_2 on X_3 , defined as $\mathbb{P}(X_3 = 1|\text{do}(X_2 = 1)) - \mathbb{P}(X_3 = 1|\text{do}(X_2 = 0))$. Appropriate bounds for this causal effect are obtained by minimizing/maximizing $q_4 - q_1 - q_2$ subject to the same constraints generated in inferences. The LCN is used only to guide the construction of an appropriate nonlinear optimization program that is not directly derived from an inference on the LCN. $\square$

Similar computations can yield probabilities for counterfactuals as defined in Pearl’s theory of causation [28]. In that theory, the probability of a counterfactual is computed by enlarging the original SCM of interest, leading to what is referred to as a *twin network*. For instance, take Example 8 and the counterfactual “ $\{X_3 = 1\}$ had X_2 been 0, given that in fact $\{X_2 = 1, X_3 = 0\}$ ”: the twin network is depicted in Figure 12. This is the network discussed previously in connection with intervention in X_2 ; we should note that the network was used

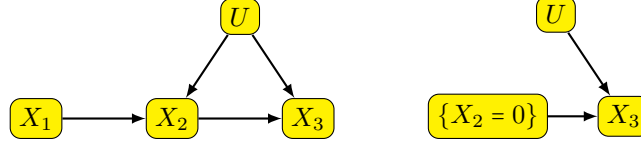


Figure 11: A structural causal model (left), and the effect of an intervention that sets the value of X_2 to 0 (right).

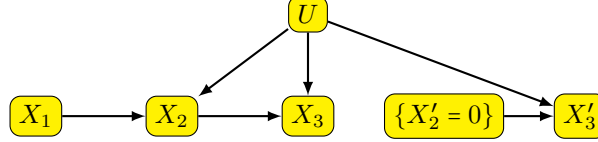


Figure 12: The “twin” network that can be used to handle interventions and counterfactuals.

there to handle probability bounds on U *without* evidence on X_1 , X_2 , X_3 , while now the twin network does have evidence in the latter variables. The whole twin network can be readily encoded into an LCN.⁷

We have so far examined a number of examples where causal reasoning can be reduced to inferences with LCNs. These examples have been dealt with linear programming [5] and credal networks [36]. But we can also consider settings that go beyond previous approaches and that require the flexibility afforded by LCNs, for instance due to cycles and due to incomplete probabilistic specifications. It seems particularly interesting to examine cyclic models as cyclic SCMs have often been discussed in the literature [6, 13, 28], but there has been scarce machinery to deal with them. So, in the next example we look at an LCN with a cycle.

Example 9. Consider an LCN whose structure is depicted in Figure 13 (left), perhaps the simplest possible SCM with a cycle and nontrivial probabilities. Suppose one writes down a few assessments and constraints:

$$\begin{aligned} \mathbb{P}(U_1 = 1) &= 0.5, & \mathbb{P}(U_2 = 1) &= 0.1, \\ X_1 &= \neg X_3 \wedge U_1, & X_2 &= X_1 \vee U_1, & X_3 &= \neg X_2 \wedge U_2. \end{aligned} \quad (10)$$

One can analyze these constraints to find, for instance, that $X_1 = U_1$; in any case, all independence relations imposed by the LMC(LCN), and also by the GMC(C), are trivially satisfied by a single probability distribution.

However, simple variants of this example lead to novel scenarios. Suppose we have:

$$\begin{aligned} \mathbb{P}(U_1 = 1) &= 0.5, & \mathbb{P}(U_2 = 1) &= 0.1, \\ X_1 &= \neg X_3 \wedge \neg U_1, & X_2 &= X_1 \vee U_1, & X_3 &= X_2 \wedge U_2. \end{aligned}$$

⁷The reader may verify that the probability of the counterfactual in the paragraph is parameterized as pt/p_{10} , with $t = \min(1, p_{01}/(1-p), p_{10}/p)$ (relevant discussion may be found in Appendix A).

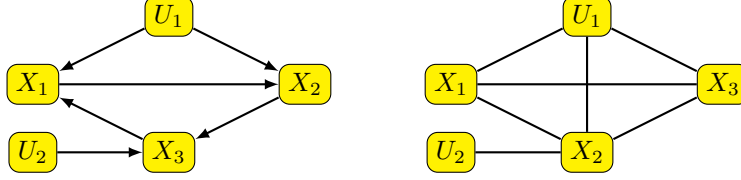


Figure 13: A structure with a cycle and the moral graph obtained from the ancestral graph of any three variables.

We now have a situation that does not occur with an acyclic directed structure: when $\{U_1 = 0, U_2 = 1\}$, we have an inconsistency (we get $X_1 = \neg X_1$). Thus a failed LCN inference should alert one that changes may be necessary to the model.

Another, perhaps more interesting, phenomenon can be contemplated if we consider an additional variant,

$$\begin{aligned} \mathbb{P}(U_1 = 1) = 0.5, \quad \mathbb{P}(U_2 = 1) = 0.1, \\ X_1 = \neg X_3 \wedge U_1, \quad X_2 = X_1 \vee \neg U_1, \quad X_3 = \neg X_2 \wedge U_2. \end{aligned} \quad (11)$$

Here we have that, when $\{U_1 = 1, U_2 = 1\}$, then $X_1 = \neg X_3$ and $X_3 = \neg X_1$. There are two interpretations that satisfy such logical constraints: either

$$\{U_1 = 1, U_2 = 1, X_1 = 1, X_2 = 1, X_3 = 0\} \text{ or } \{U_1 = 1, U_2 = 1, X_1 = 0, X_2 = 0, X_3 = 1\}.$$

Any joint probability distribution that takes probability mass $\mathbb{P}(U_1 = 1, U_2 = 1)$ and distributes it between these two interpretations is a valid distribution. Thus we obtain a set of probability distributions as the semantics of the LCN, leading for instance to an interval-valued inference $\mathbb{P}(X_2 = 1) \in [0.95, 1]$. And in this example, an intervention on X_1 “breaks” the cycle and leads to a precise interventional inference $\mathbb{P}(X_2 = 1 | \text{do}(X_1 = 0)) = 0.5$. $\square$

A point to emphasize concerning this example: if one is to handle cycles, one must adopt a formalism that can deal with inconsistency and with probabilistic imprecision from the outset. Note that similar scenarios can be found in probabilistic logic programming [32], where a program with cycles may have more than one model per realization of probabilistic variables (there called *facts*). The *credal semantics* for such cyclic programs may then interpret a probabilistic logic program through a *set* of probability distributions.

Another point is the interplay between SMCs and Markov conditions. Note that, in the previous example, the LMC(LCN) is satisfied because the root nodes are independent and the other nodes are deterministically determined by their parents. Thus one might even learn the probabilities of exogenous variables (the root nodes) from data, for this cyclic model, without a concern over the fact that some conditional probabilities are zero due to constraints, as long as the LMC(LCN) is adopted. In fact, satisfaction of local Markov conditions is guaranteed for any SCM-based LCN simply due to the fact that endogenous

variables are deterministically determined by their parents. To compare, for the GMC(C) a factorization is not guaranteed when probabilities may be equal to zero, and in fact global Markov conditions may fail if one tries to use them with a SMC [27]. This may be an interesting point through which Markov conditions may be compared.

To investigate further differences between Markov conditions, we may look for an example where the LMC(LCN) may fail to produce a distribution and yet the GMC(C) over the structure does not. We must then move away from SCMs for moment, and also we must focus on cycles, if we wish to find a challenge to the LMC(LCN). The next example does so by using a cyclic causal diagram.

A causal diagram is a directed graph where each node is a variable and where edges are interpreted causally [29]. Their semantics can be based on functional relationships between variables, their parents, and additional variables not drawn in the graph and that work as disturbances. For our purposes, we do not need to dwell into such disturbance; we can instead suppose that each variable X is associated with probability distributions for X given its parents in the graph. Interventional and counterfactual queries are still possible by cutting causal diagrams appropriately [28]. Usually a causal diagram is assumed to have no cycles, but practical circumstances may introduce feedback or mutual constraints that lead to cycles [13]. This is the scenario we wish to explore.

Example 10. Suppose one specifies an LCN as follows:

$$\begin{aligned}
& \mathbb{P}(U = 1) = 0.5, \\
& \mathbb{P}(X_1 = 1|U = 0, X_3 = 0) = 0.1, \quad \mathbb{P}(X_1 = 1|U = 0, X_3 = 1) = 0.3, \\
& \mathbb{P}(X_1 = 1|U = 1, X_3 = 0) = 0.7, \quad \mathbb{P}(X_1 = 1|U = 1, X_3 = 1) = 0.9, \\
& \mathbb{P}(X_2 = 1|U = 0, X_1 = 0) = 0.8, \quad \mathbb{P}(X_2 = 1|U = 0, X_1 = 1) = 0.3, \\
& \mathbb{P}(X_2 = 1|U = 1, X_1 = 0) = 0.7, \quad \mathbb{P}(X_2 = 1|U = 1, X_1 = 1) = 0.4, \\
& \mathbb{P}(X_3 = 1|T = 1) = 0.2, \quad \mathbb{P}(X_3 = 1|X_2 = 0) = 0.7.
\end{aligned} \tag{12}$$

The structure is, of course, similar to the one in Figure 13, now without node U_2 . The local Markov condition LMC(LCN) only imposes $X_3 \perp\!\!\!\perp U|X_2$. Alas, no joint distribution can satisfy the constraints and this independence relations — we have a failure of probabilistic coherence, rather than of logical consistency. However, note that the global Markov condition GMC(C) imposes *no* independence relation, and one can find many distributions that satisfy the constraints. Here is a situation where the global Markov condition is in fact less stringent than the local one. $\square$

The previous examples are perhaps too small, but with a few additional nodes we already reach a structure, depicted in Figure 14, that has been actually advocated as a model of the popular hippocampus dataset on brain activity [14]. We use the structure in Figure 14 the next example. In addition, the next example deals with interval-valued assessments: as noted at the beginning of this section, in practice one may face additional sources of imprecision in probability values, as alluded previously, perhaps due to limited understanding of the domain, disparate opinions of experts, incomplete experiments.

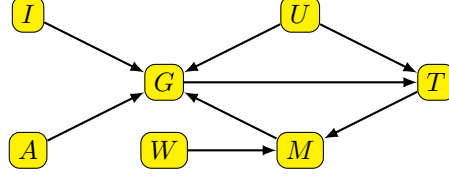


Figure 14: Cyclic diagram.

Example 11. Suppose a person is discussing greenhouse gases using the following sketchy model. There are variables that denote increase in Industrial activity (I), in Agricultural activity (A), in Greenhouse gases (G), in Temperature (T), and in permafrost Melt (M). There are also two exogenous variables U and W (U is a confounder that influences both G and T). For all those variables, value 1 denotes increase, and 0 otherwise. And suppose we have assessments that again imply the structure in Figure 14:

$$\begin{array}{ll}
0.85 \leq \mathbb{P}(I = 1) \leq 0.9, & 0.2 \leq \mathbb{P}(A = 1) \leq 0.25, \\
0.5 \leq \mathbb{P}(G = 1 | \{I = 1\} \wedge \{A = 1\}) \leq 0.80, & 0.7 \leq \mathbb{P}(T = 1 | G = 1) \leq 0.75, \\
0.75 \leq \mathbb{P}(M = 1 | T = 1) \leq 0.8, & 0.15 \leq \mathbb{P}(G = 1 | M = 1) \leq 0.2, \\
0.1 \leq \mathbb{P}(G = 1 | U = 0) \leq 0.15, & 0.85 \leq \mathbb{P}(G = 1 | U = 1) \leq 0.9, \\
0.45 \leq \mathbb{P}(T = 1 | U = 0) \leq 0.5, & 0.65 \leq \mathbb{P}(T = 1 | U = 1) \leq 0.7, \\
0.1 \leq \mathbb{P}(M = 1 | W = 0) \leq 0.5, & 0.5 \leq \mathbb{P}(M = 1 | W = 1) \leq 0.9.
\end{array}$$

An LCN can be coded with these assessments so as to obtain inferences; if the LMC(LCN) is adopted, it is necessary to enforce independence relations such as $A \perp\!\!\!\perp I, U, W$ and $T \perp\!\!\!\perp A, I, W | G, U$. A twin network, built using Pearl's analysis of counterfactuals [28], can also be expressed through an LCN, hence existing inference algorithms [23] can be used in counterfactual reasoning.

For instance, the bound of the unobserved marginal of U is $\mathbb{P}(U = 1) \in [0.0, 0.26]$. As another example, we have $\mathbb{P}(T = 1 | \text{do}(G = 1)) \in [0.0, 0.96]$. Now consider the counterfactual query,

Would temperature increase, had greenhouse gases been increased, given that in fact decreasing greenhouse gases have increased temperature?

An answer to this question, we have $\mathbb{P}(T_{\text{do}(G=1)} = 1 | G = 0, T = 1) \in [0.50, 0.92]$.

In order to further explore this example, suppose one adopts the GMC(C) over the structure. To obtain the resulting independence relations, note that any subset of variables that contain either G or T or M automatically brings these three variables, and in fact all others, as ancestors into the ancestral moral graph. Hence we have two situations: either we study independence relations between variables that are neither G nor T nor M , and from there we obtain that A and I and U and W are independent, or we study independence relations that are obtained by separation in the moral graph of the whole structure. This moral graph is depicted in Figure 15. The cliques are $\{A, I, G, M, U\}$,

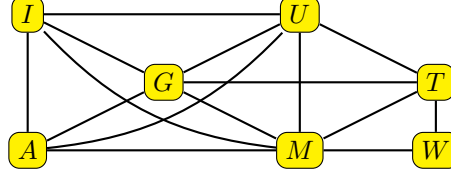


Figure 15: Moral graph of the directed graph in Figure 14.

$\{G, M, T, U\}$ and $\{M, T, W\}$ (the first two are separated by $\{G, M, U\}$, while the latter two are separated by $\{M, T\}$). There are thus two independence conditions extracted from this moral graph; namely, $A, I \perp\!\!\!\perp T, W | G, U, M$ and $W \perp\!\!\!\perp A, U, I, G | M, T$. The factorization property thus yields a joint probability distribution that is proportional to $f_1(A, I, G, M, U)f_2(G, M, T, U)f_3(G, M, U)$, where these three functions f_1 , f_2 and f_3 are positive. Now this factorization is valid if the probability of each complete configuration of variables is positive, something that cannot be guaranteed with the previous assessments. One would have to impose some additional condition, say that every configuration has probability larger than or equal to some small quantity.

□

8. Conclusion

In this paper we visited many Markov conditions that can be applied, if properly adapted, to Logical Credal Networks [22]. We reviewed existing concepts and introduced the notion of structure of an LCN, showing that the original local condition $\text{LMC}(\text{LCN})$ can be viewed as a local condition on structures. We then showed that the $\text{LMC}(\text{LCN})$ is equivalent to a usual local condition when the structure is a chain graph, and this leads to a factorization result. Moreover, we presented a new semantics based on structures and a global Markov condition — a semantics that agrees with the original one when the structure is a chain graph but that offers a possible path to factorization properties. We also investigated the potential of LCNs as modeling and inferential tools in causal reasoning, a topic that necessarily deals with probability bounds due to non-identifiability.

There are many issues left for future work. LCNs stress the connection between the syntactic form of constraints and the semantic consequences of independence assumptions, a theme that surfaces in many probabilistic logics. We must investigate more carefully the alternatives when extracting independence relations from constraints, in particular to differentiate ways in which bi-directed edges are created.

We must also examine positivity assumptions. What is the best way to guarantee a factorization? Should we require the user to explicitly express positivity assumptions? Should we allow for logical constraints that assign probability zero

to some configurations; if so, which kinds of configurations? How to make such constraint-induced zero probabilities compatible with factorization properties?

It is also important to study a large number of Markov conditions that can be found in the literature but that we have skipped here, both conditions connected with chain graphs and conditions connected with causal and feedback models. We must verify which conditions lead to factorization results, and which conditions are best suited to capture the content of logical formulas, causal influences, feedback loops.

In a more applied perspective, it seems profitable to explore in more detail the connection between LCNs and causal reasoning. There are practical issues (how best to compute the causal effect of an intervention coded through an LCN?) and theoretical ones (can all feedback models be captured by LCNs?). It is also important to investigate whether the ideas behind LCNs can be used with practical specification languages such as Probabilistic Answer Set Programming, and we must test how various semantics for LCNs fare in realistic settings.

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Appendix A. Probability Bounds in Example 8

Here we present detailed symbolic expressions for quantities discussed in Example 8. Denote $\mathbb{P}(U = u_i)$ by q_i and recall that $\mathbb{P}(X_2 = j, X_3 = k)$ is denoted by p_{jk} (thus $p_{jk} = p_{0jk} + p_{1jk}$) and $\mathbb{P}(X_1 = 1)$ is denoted by p (thus $p = \sum_{jk} p_{1jk}$). By analyzing the structural equations, we have $q_0 = p_{00}$ and $q_4 = p_{11}$; moreover,

$$p_{01} = \mathbb{P}(U = u_1) + \mathbb{P}(X_1 = 0, U = u_2) = q_1 + (1 - p)q_2,$$

where the last term is obtained by independence of X_1 and U . Likewise,

$$p_{10} = \mathbb{P}(U = u_3) + \mathbb{P}(X_1 = 1, U = u_2) = q_3 + pq_2.$$

Thus we can parameterize the vector $[q_1, q_2, q_3, q_4, q_5]$ as

$$[p_{00}, p_{01} - (1 - p)t, t, p_{10} - pt, p_{11}],$$

with $t \in [0, \min(1, p_{01}/(1 - p), p_{10}/p)]$.

Despite the fact that the exogenous variable is associated with a set of marginal probability distributions, the conditional probability of X_3 given X_2 is, by construction, a precise number. For instance, we have $\mathbb{P}(X_3 = 1 | X_2 = 0) = p_{01}/(p_{00} + p_{01})$. However, consider an intervention that sets X_2 to 0; what is then the probability of X_3 under such intervention? Take the sub-graph obtained by

removing all edges into X_2 . This process leaves X_1 isolated, so we do not draw it in Figure 11 (right). In this figure we explicitly indicate that X_2 is set to 0.

The effect of the intervention is then:

$$\begin{aligned}\mathbb{P}(X = 1 | \text{do}(X_2 = 0)) &= \sum_{i=0}^4 \mathbb{P}(X_3 = 1 | X_2 = 0, U = u_i) \mathbb{P}(U = u_i) \\ &= q_1 + q_2 = p_{01} - (1 - p)t + t \\ &= p_{01} + pt, \quad \text{for } t \in [0, \min(1, p_{01}/(1 - p), p_{10}/p)],\end{aligned}$$

as $\mathbb{P}(X_3 = 1 | X_2 = 0, U = u_i) = 1$ for $i = 2, 3$ and 0 otherwise. Clearly the value of $\mathbb{P}(X = 1 | \text{do}(X_2 = 0))$ can vary within a non-trivial interval (given the assumption that all observed probabilities are positive).

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